최적궤도제어

기말고사 2014 년 가을

1. Find optimal control profile in terms of the costates.

비교하기 transversality condition
~~유도~~ 조건에서 $\vec{\lambda}$ 구하기
→ (1) 최소화하는 u 구하기

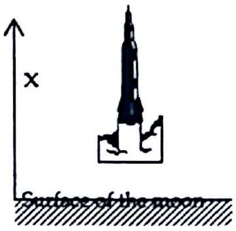
The state equations are
$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -x_2(t) + u(t)$$

The performance index is $J = \int_0^{t_f} \frac{1}{2}[x_1^2(t) + u^2(t)]dt$

Initial states are given, final time is specified, and the final state are free,
The constraint on control is $-1 \le u(t) \le 1$ for all time.

$|\vec{u}(t)| < 1$

2. A lunar rocket in the terminal phase of a minimum-time will make a soft landing on the surface of the moon.



m ; spacecraft mass
$-M \le \dot{m} \le 0$; spacecraft mass rate, control variable
g ; gravitational constant in surface of the moon
T ; thrust ($T = -k\dot{m}$)
where k is constant relative exhaust velocity of gas

$x_1 = x$
$x_2 = \dot{x}$
$x_3 = m$
$x_4 = u$

The equation of motion is following.
$$m(t)\ddot{x}(t) = -gm(t) + T(t)$$

2-1) Solve the problem using Pontryagin's minimum principle.
2-2) Determine whether or not there can be any singular intervals.

↳ for a finite time, $(\ )\vec{u}$ term $= 0$ 이라고 보고 가능한 구간 있는지 보기.

3. The system equation and performance index are
$$\dot{x} = u, \qquad x(0) = x_0 \qquad x(1) = 0$$
$$J = \int_0^1 g(x,u)dt$$

Derive the necessary conditions for optimality if x is suddenly changed at a time $0 < t_1 < 1$ as follows; $x(t_1^+) = x(t_1^-) + 1$
Is the Hamiltonian continuous or not?

4. Explain how to solve numerically the following optimal control problems by using a direct method, as detail as possible.

→ 자세하게!

$$\dot{x}(t) = -x(t) + u(t), \qquad x(0)=4$$

It is desired to find $u(t)$, $t \in [0,1]$. The performance index is
$$J = x^2(1) + \int_0^1 \frac{1}{2}u^2(t)dt$$

AST 6037-01 Trajectory Optimization

Final Examination

December 11, 2012 10:00-12:00 (120 minute)

Close Books/Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) Consider minimizing the following functional

$$J(x,\dot{x}) = \int_{t_0}^{t_f} L(x,\dot{x},t)dt$$

where $x(t_0)$ and $x(t_f)$ are free.

I. (10) Use the fundamental theorem of the Calculus of Variations to derive the necessary conditions for an extremum.

II. (10) Determine an extremum for $L(x,\dot{x},t) = \frac{1}{2}\dot{x}^2 + x\dot{x} + \dot{x} + x$, $t_0 = 0$ and $t_f = 2$

2. (20) Consider minimizing the performance index

$$J = h\big(x(t_f),t_f\big) + \int_{t_0}^{t_f} g\big(x(t),u(t),t\big)dt$$

subject to the dynamical equations of motion

$$\dot{x}(t) = f(x,u,t)$$

while satisfying the boundary conditions

$$x(t_0) = x_0, \quad m(x(t_f),t_f) = 0$$

where $x \in R^n$, $u \in R^m$, $m \in R^k$, $m \le n$, $k \le n$ and t_f is free. Use the variational principle to derive the first order necessary conditions for optimality.

3. (30) Consider minimizing the performance index

$$J = \int_0^2 (u-1)x dt$$

subject to the dynamics

$$\dot{x} = xu, \quad x(0) = x_0, \quad x > 0$$

and control bounds

$$0 \leq u \leq 1$$

I. (10) Use the Pontryagin Principle to derive the first order necessary conditions for optimality.

II. (10) Show that there does not exist any singular interval.

III. (10) Develop a candidate optimal trajectory explicitly for state, costate and control.

4. (10) Consider minimizing the performance index

$$J = \int_{t_0}^{t_f} |u(t)|\, dt$$

subject to the dynamics

$$\dot{x}(t) = u(t),\ \ x(t_0) = x_0,\ \ x(t_f) = 0,\ \ t_f \text{ is free}$$

and control bounds

$$|u(t)| \leq 1.0$$

The control objective is for the system to transfer from an arbitrary initial state x_0 to the origin, while minimizing fuel consumption. Use the Pontryagin principle to determine the control logic.

5. (20) Given a function $f : R^n \to R$, suppose that its gradient $G(x_k) \neq 0$ and Hessian $H(x_k) > 0$ at x_k. Show that the direction

$$d_k = -H^{-1}(x_k) G^T(x_k)$$

is a descent direction. That is, show that if

$$x_{k+1} = x_k + \alpha_k d_k$$

then

$$f(x_{k+1}) < f(x_k)$$

for sufficiently small $\alpha_k > 0$.

최적궤도제어

기말고사 20011 년 가을

1. $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} + \mathbf{f}(t)$, $\mathbf{x}(0) = \mathbf{x}_0$

f(t) is a known forcing function.

$$J = \frac{1}{2}(\mathbf{x}^T Q\mathbf{x})_{t=t_f} + \frac{1}{2}\int_0^{t_f} (\mathbf{x}^T Q\mathbf{x} + \mathbf{u}^T R\mathbf{u})dt$$

t_f is known. Show that

$$\mathbf{u}(t) = -R^{-1}B^T(k\mathbf{x} + \mathbf{r})$$

$$\dot{\mathbf{r}}(t) = (kBR^{-1}B^T - A^T)\mathbf{r} - k\mathbf{f}(t)$$

k satisfies the Riccati equation.

2.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -ax_2 - bx_1 + u$$

$$J = \frac{1}{2}\int_0^{t_f} [t(c_1x_1^2 + c_2x_2^2)]dt$$

where a, b, c_1 and c_2 are constant. Find the optimal control law on singular intervals and on regular intervals.

3. The system equation are

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = u_1$$

$$\dot{x}_3 = x_4, \qquad \dot{x}_4 = u_2$$

$\mathbf{x}(0)$ are given and $\mathbf{x}(t_f)$ are free.

There is a state equality constraint as:

$S(\mathbf{x}) = x_1^2 + x_3^2 - 1 = 0$ $\to \dot{S} = 0$ $+ C(\vec{x}, \vec{u}, t) = 0$

The performance index to be minimizes is

$J = \int_0^{t_f} (u_1^2 + u_2^2)dt$, t_f is fixed →

augment

H 구하기

→ optimality condition 에서 u_1, u_2 구하기

→ C = 0 에 대입하면 μ 구해짐

$\dot{\lambda} = -\frac{\partial H}{\partial x}$

Explain how to solve this problem.

$S(\vec{x}(t_f)) = 0$ $\vec{\lambda}(t_f) = \vec{\nu}^T \frac{\partial S}{\partial \vec{x}}\Big|_{t=t_f}$

4.

Explain the algorithm of a <u>direct method</u> and explain also the algorithm of an <u>indirect method</u>, in order to numerically solve optimal control problems.

최적궤도제어

기말고사　　**2017 년 가을학기**

1. (25 points)
   Derive and explain the Pontryagin's Minimum Principle.

2. (25 points)
   The system $\dot{x}(t)=u(t)$
   is to be transferred from an arbitrary initial state x_0 to the origin. The performance index to be minimized is

$$J=\int_0^{t_f}|u(t)|dt$$

$\Rightarrow 1$　C, C-1

Where t_f is free, and the admissible control law satisfy $|u(t)|\le 1$.

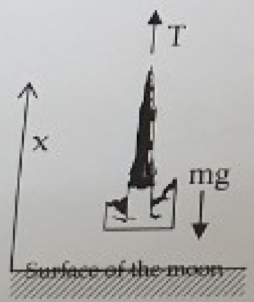
2-1)　Determine the optimal control law.

2-2)　Show that $x_0=-\int_0^{t_f}u(t)dt$　for $x(t_f)=0$.

2-3)　Suppose that $x_0=5$, and show that each of the following controls satisfies the equation in 1-2) and makes $J=5.0$. What are the implications of this result.

$u(t) = -1,\quad t\in[0, 5]$
$u(t) = -0.5,\quad t\in[0, 10]$
$u(t) = -0.2,\quad t\in[0, 25]$
$u(t) = -0.1,\quad t\in[0, 50]$

3. (30 points)
   The figure shows lunar rocket in the terminal phase will make a soft landing on the surface of the moon.



m　; spacecraft mass
$-M\le\dot{m}\le 0$　; spacecraft mass rate, control variable
g　; gravitational constant in surface of the moon
T　; thrust ($T=-k\dot{m}$) >0
where k is constant relative exhaust velocity of gas

3-1)　Show the equation of motion (1-demension)

3-2)　Define the states of the system and control and show the state equations.

3-3)　Determine the costate equations and boundary conditions for the soft lunar landing vehicle to be accomplished with minimum-time.

3-4)　Determine necessary conditions for the landing to be accomplished with minimal-fuel expenditure.

3-5)　Using the following differential equation